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Health Insurance Cost Prediction Using Regression Models

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***Abstract*: India's government spends 1.5 percent of its annual GDP on public healthcare, which is significantly less than that of other countries. Global public health spending, on the other hand, has almost doubled in line with inflation in the last two decades, reaching US$ 8.5 trillion in 2019, or 9.8% of global GDP. Multinational multi-private sectors provide around 60% of comprehensive medical treatments and 70% of out-patient care, which charge patients astronomically high fees. Because of the rising expense of quality healthcare, increased life expectancy, and the epidemiological shift toward non-communicable diseases, health insurance is becoming an essential commodity for everyone. Insurance data has increased dramatically in the last decade, and carriers now have access to it. The health insurance system explores predictive modeling to boost its business operations and services. Computer algorithms and Machine Learning (ML) is used to study and analyze the past insurance data and predict new output values based on trends in customer behavior, insurance policies, and data-driven business decisions, and support in formulating new schemes. Additionally, ML has found enormous and potential applications in the insurance industry. Thus, this paper develops a real-time insurance cost price prediction system named ML Health Insurance Prediction System (MLHIPS) using ML algorithms which will aid the insurance companies in the market for easy and rapid determination of values of premiums and thereby curb down health expenditure. The proposed model incorporates and demonstrates different models of regression such as Ridge Regression, Lasso Regression, Simple Linear Regression, Multiple Linear Regression and Polynomial Regression to anticipate insurance costs and assess model outcomes. In the proposed model, the Polynomial Regression model has achieved better results with an RMSE value of 5100.53 and R -squared value of 0.80 compared to all the other models.**

Keywords— Machine Learning, Regression Models, Ridge Regression, Lasso Regression, Linear Regression, Multiple Linear Regression and Polynomial Regression.

I. INTRODUCTION

Public health is an integral part of the society, of the country and of the world we live in and is an important matter of concern. Human lives and public health may be endangered by natural calamities, global epidemic and pandemics, global crisis of medical aids, etc. which increases the vulnerability of public health. People can meet with unavoidable and unforeseen circumstances at any point of their lifetime. Individuals, families, companies and properties are uncovered and uninsured to diverse hazard forms, natural calamities, and the likelihood can shift. These perils include the possibility of mortality, health, and property disaster or resource depletion.

People's lives revolve around two main elements: life and prosperity. However, keeping a safe and sound distance from unforeseen events is impossible. The Government of India spends 1.5% of GDP on public health, the lowest level globally [1]. In the last two decades, universal health care has been almost doubled, surpassing US $ 8.5 trillion in 2019, or 9.8% of global GDP [2]. The multi-private sector having state-of-the-art facilities provides almost 60% of total hospitalizations and 70% outpatient services [3]. Thus, Health insurance is becoming an essential commodity for every individual because of the increasing cost of quality healthcare combined with higher life expectancy and widespread transition towards non-communicable diseases. To alleviate these types of problems, the world of fund has developed a variety of tools to protect people and organizations from such unseen catastrophic situations, through the utilization of monetary capital to repay and compensate them. In this manner, insurance is an arrangement that diminishes or evacuates misfortune costs brought about by different dangers.

Today, data has evolved drastically since the recent past decade and insurance carriers have access to it. The health insurance system is exploring ways to use predictive modelling to boost their business operations and services. Computer algorithms and Machine Learning (ML) are used to study and analyze the historical insurance data and predict new output values based on trends in customer behavior, insurance policies and data-driven business decisions, and supports in formulation of new schemes. Besides, most

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insurance companies use conventional databases to store their data which is primarily structured data. Moreover, merely 10-15 percent of the total data available is processed for gaining insights. Thus, transformation of the data is necessary to gain valuable insights that may be very crucial for the growth of such companies. Therefore, analyzing the structured data as unstructured data for making better decisions requires statistical and Machine Learning (ML) techniques. The main advantage of ML is that it can be effectively applied to a massive volume of structured, semi-structured, or unstructured datasets. The ML model can be used across multiple value chains to understand the weight-age of risk involved, claims made and customer behavior with greater predictive accuracy. ML applications in the health insurance sector include various tasks such as understanding risk tolerance and premium leakage leading to inaccurately pricing of premiums, loss deterrence, claims handling, expense management, subrogation, litigation, and fraud identification.

When it comes to the value of health insurance in people's lives, it's vital for insurance firms to be as explicit as possible for measuring the sum secured by this approach and the protection charges which must be paid for it. The calculation of health insurance charges in the traditional process are a hefty task for the insurance companies. The intervention of humans in this process may sometime produce faulty or inaccurate results. Additionally, as the data increases manual calculations becomes lethargic and time consuming. Again, in such scenarios the implementation of ML models can be very beneficial for such companies. Therefore, ML may generalize the exertion or strategy to define such an approach. These models can perform self-learning to predict the cost of insurance using past insurance data of the companies. The model inputs are the main parameters that are utilized to calculate the instalments made. This enables the algorithm to precisely estimate the disbursement of insurance coverage. In this way, the correctness can be progressed with ML. The objective of the proposed model is to perform rapid estimation and prediction of insurance charges at a hospital incurred by a patient, using ML models upon the Kaggle dataset. Thus, this paper develops a real-time insurance cost price prediction system named ML Health Insurance Prediction System (MLHIPS) using ML algorithms which will aid the insurance companies in the market for easy and rapid determination of values of premiums and thereby curb down health expenditure. The proposed model incorporates and demonstrates different regression models such as Multiple Linear Regression, Ridge Regression, Simple Linear Regression, Lasso Regression and Polynomial Regression to predict the insurance costs and compares the models based on their results.

II. LITERATURE S URVEY

Several studies on estimating medical prices have been published in the health field in different contexts [4], [5], [6], [7], [8]. Machine learning has many probable assumptions, but its performance relies on picking a nearly precise algorithm for the specified problem domain and following the appropriate procedures to build, train, and deploy the model.

Moran et al. [4] "utilized a comprehensive linear regression method which was used to predict the cost of an intensive care unit (ICU) using patient profile data, DRGs (diagnosis-related groups), the length of time spent in the hospital, and additional traits as features."

Sushmita et al. [5] "proposed a model based on the medical and past expense history of a person to predict his/her future medical costs. Quarterly projected expenses for the future 3,6,9 and 12 months were estimated with the use of the model. They have used random forest and linear regression analysis to predict the costs."

Lahiri et al. [6] "used a classification algorithm to predict whether an individual's medical expenses would increase in the next year, taking into account the medical expenses of the previous year."

Gregori et al. [7] have used the logit model and the OLS method to study the multivariate modelling of healthcare costs data.

Bertsimas et al. [8] use data mining techniques, explicitly clustering algorithms and classification trees, and insurance claim data of nearly 500,000 members throughout a three-year period. Based on the data gathered from the medical expenses from the first two years, a justified third-year health-care cost projection is made.

III. METHODOLOGY

The regression techniques used are the statistical methods that establishes the association between a target or dependent variable and a set of independent or predictor variables. It assumes that both the target and the predictor variables are having numerical values and there exists some kind of corelation between the two. The models that we are implementing in our problem are discussed below,

A. Model Selection:

1) *Simple Linear Regression:* In simple linear regression [16], the target variable(Y) is dependent on a single independent variable(X) and the model establishes a linear relationship among these two variables.

The linear regression model tries to fit the regressor line between the independent(X) and dependent(y) variable. The equation of the line is given by:

$$Y = a + bX \quad (1)$$

where "a" and "b" are the models parameters called as regression coefficients, "a" is the value of the Y intercept that the line makes when X is equal to zero and "b" is the slope that signifies the change of Y with the change of X. More the value of "b" means a small change in X causes a significant change in Y, and vice versa. The value of "a" and "b" can be found by Ordinary Least Square method.

In linear regression models the values predicted may not be accurate always, there will always be some difference

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hence we add an error term ϵ to the original equation (1) that accounts for the difference and thus help in making better predictions.

$$Y = a + bX + \epsilon \quad (2)$$

Assumptions in Linear Regression

- The sample size of data should exceed the number of available parameters.
- Only over a restricted range of data the regression can be valid.
- Error term is normally distributed. This also means that the mean of the error has expected value of 0.

2) *Multiple Linear Regression:* Similar to simple linear regression, multiple regression [17] is a statistical procedure that examines the degree of association between a set of independent variables and a dependent variable. There is just one independent and one dependent variable in basic linear regression, but there are numerous predictor variables in multiple linear regression and the value of dependent variable(Y) is now calculated depending on the values of the predictor variables. it is assumed that there is no dependency among the predictor variables. Suppose if the target value is dependent on "n" independent variables then the regressor fits the regression line in a N dimensional space. The regressor line equation is now modified into

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 + \ldots.. + b_nX_n + \epsilon \quad (3)$$

where "a" is the Y-intercept value and < b_1, b_2, b_3, …., bn > are the regression coefficients associated with the n independent variables and ϵ is the error term.

3) *Polynomial Regression:* Polynomial Regression [18] is yet another a special case of linear regression. In Linear regression the model tries to fit a straight regression line between the dependent and independent variable. In scenarios where there doesn't exist linear relationship between the target and predictor variable then instead of a straight line a curve is being fitted against the two variables. This is accomplished by fitting a polynomial equation of degree n on the non-linear data which establishes a curvilinear relationship among the dependent and independent variables. In polynomial regression the assumption that the independent variables must be independent of each other is not mandatory. The equation of the line thus reduces to:

$$Y = a + b_1X^1 + b_2X^2 + b_3X^3 + \ldots. + b_nX^n + \epsilon \quad (4)$$

The following are some of the benefits of applying polynomial regression:

- Polynomial Regression offers the best estimate of the relationship between the dependent and independent variable.
- Higher degree polynomial generally provides a good fit on the dataset.
- Polynomial Regression basically fits a wide range of curves of varying degree to the dataset.

Drawbacks of applying Polynomial Regression:

- These are too sensitive to the existence of outliers in the dataset, as outliers cause the model's variance to rise. When the model comes across an unknown data item, it underperforms.

4) . *Ridge Regression:* Ridge regression [19] is a standard model tuning process used to analyse the data suffering from multicollinearity. This is a way to approximate the coefficients of the regression model when the independent variables are firmly related or there exists an association between them. Ridge Regression's main objective is to take the dataset and fit a new line into it in a way that does not overfit the model. For this purpose, ridge regression adds an insignificant amount of bias that determines the fitting of the line into the data. We obtain a substantial reduction in variance, which leads to an increase in the accuracy value by the addition of bias. The least Square determines the values of the parameters for the equation (1), which diminishes the sum of squared residuals. But in contrast the Ridge Regression regulates the value for parameters that results in minimization of the sum of squared residuals along with an additional term λ*b2. Ridge Regression performs L2 regularization.

$$Loss_{Ridge} = \sum(y_i - y^*)^2 + \lambda b^2 \quad (5)$$

where y* = a + bX is the predicted value.

In this ridge regression method, the coefficients are penalised by a value lambda, this acts as a control parameter, which determines how severe is the penalty and how much significance should be given to Xi. The higher the values of the lambda the bigger is the penalty and therefore the magnitude of coefficients is reduced.

When the slope (b) of the line is steep then the target variable(Y) is very sensitive to relatively small change in the predictor variable variable(X). In ridge regression by the addition of the lambda value the sensitivity decreases. If lambda is zero then ridge regression reduces to linear regression and when lambda increases gradually the slope the line decreases asymptotically. To know which value of lambda is to choose we try different values of lambda and use cross validation to determine which one result in the lowest variance.

5) *Lasso Regression:* Least Absolute Shrinkage and Selection Operator (LASSO) [20] is similar to ridge regression that penalizes the regression model. It performs L1 regularization. The Lasso regression process is usually used in machine learning for the selection of the significant subset of variables. The prediction accuracy of this model is higher when compared to other model interpretations. Like ridge regression lasso regression also results in a line with little amount of bias added to it which thereby decreases the variance of the model.

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$$Loss_{Lasso} = \sum(y_i - y^*)^2 + \lambda|b| \quad (6)$$

The major difference between lasso and ridge regression is, ridge regression decreases the slope asymptotically close to zero but lasso regression can reduce the slope all the way down to zero, thereby eliminating the useless parameters from the line equation that don't have any significant role for predicting the value of the target variable.

Lasso regression usually works better under conditions where some predictors have high coefficients, and the rest have low coefficients. Ridge regression performs better when the result is a function of many predictors, all of which have coefficients of approximately the same size.

B. Description of the Dataset:

From the Kaggle site [21]we obtained our dataset for developing the ML Health Insurance Prediction System (MLHIPS). The data set obtained contains seven attributes or features and 1338 rows; out of the seven attributes or features three of them contains categorical values and the rest contain numerical values. The data set is then divided into two halves. The first component is referred to as training data, while the second is referred to as test data. The more data that is supplied to the model during its training period, the more accurate the model will be when making predictions on unseen data. Data is typically split at a ratio of 80:20 for testing and training purposes. Training datasets are used to build models as predictors of health insurance costs, and test sets are used to evaluate regression models. Table.1 displays the dataset's description.

The dataset contained missing values in certain fields. After reviewing the distributions, it was decided to replace the missing variables with new attributes, implying that the data is missing. [9]. This is only possible if the data is lost completely at random; thus, the missing data mechanism, which determines the best method to data processing, must first be developed. [10] [11]. Multilevel structure and hidden dependencies are present in medical data [12]. It is vital to figure out these hidden patterns and use various fundamental analysis techniques present in a combined fashion. This is the reason why in the field of medical data analysis, many researchers use different ensemble Machine Learning models. In order to tackle the problem of price prediction, researchers have also used hierarchical regression analysis. Many of them have used different ensemble learning techniques such as Random Forests, Adaboost, GBM, and XGBM. In paper [13], to forecast the modulus of elasticity of the collective recycled concrete, an ensemble of Random Forests (RF) and Support Vector Machines (SVM) is utilised. This classical ensemble has a much higher level of precision. To take advantage of the heterogeneity of distinct sets of meta-features, an ensemble of K-Nearest Neighbor (KNN) classifiers was created for recommendation purposes is analysed in [14].

For the diabetic retinopathy dataset, researchers used an ensemble-based machine learning model that included ID3, Random Forests, Adaboost, Logistic Regression and KNN, [15].

C. Data Pre-processing:

The dataset contains seven variables, as shown in the table above. While calculating the cost of the Charges of a customer which is our target variable the values of the rest six of the variables are taken into consideration. In this phase, the data is reviewed, properly reconstructed, and properly applied to machine learning algorithms. The dataset was first checked for missing values. The dataset was found containing missing values in the bmi and charges columns. The missing values were imputed by the mean values of the respective attribute values.

As regression models accept only numerical data, the categorical columns in our case the sex, smoker and region columns containing categorical columns were converted into numerical values using label encoding. Then the updated dataset was partitioned into training and testing dataset. And the model was trained using the training dataset.

TABLE I
DATASET

Name	Description
Age	Customer's Age
BMI	Body mass index of the customer
Number of kids	Number of kids of the customer
Gender	Male / Female
Smoker	Whether the customer is smoker or not.
Region	Where the customer lives: southwest, southeast, northeast, northwest
Charges (target variable)	Medical fee the customer has to pay

IV. RESULTS

The regression model's performance is evaluated on the basis of the following metrices

- R^2_Score
- Root Mean Square Error (RMSE)

R^2_Score: R-Squared is a good measure to evaluate the model fitness. The R-squared value lies between 0 to 1 (0% to 100%). Large value represents a better fit.

$$R^2 = 1 - \frac{SSE}{SST} \quad (7)$$

where SSE (Squared sum of error): sum of the squared residuals, which is squared differences of each observation from the predicted value. $\sum(y_i - y^*)^2$ and,

SST (Sum of Squared Total): squared differences of each observation from the overall mean. $\sum(y_i - \hat{y})^2$, where

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y_i represents the observed value, y^* is the projected value and $\hat{y}$ is the average of the observed values.

RMSE: The Root Mean Square error is a common method of calculating a model's prediction error. The RMSE, which represents how close the observed data points are to the model's predicted values, shows the model's absolute fit to the data points. A better match is indicated by lower RMSE values

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(y_i - y^*)^2}{n}} \quad (8)$$

TABLE II. CALCULATED RESULTS

Regressor	R2_Score	RMSE	Accuracy (in %)
Simple Linear Regression	0.62	7523.98	62.86
Multiple Linear Regression	0.75	7523.98	75.86
Polynomial Regression	0.80	5100.53	80.97
Ridge Regression	0.75	6070.80	75.82
Lasso Regression	0.75	6066.31	75.86

The outcomes of the above discussed models after testing the models on the test dataset were noted. Table.2 displays the R^2, RMSE and Accuracy metrices of the models discussed so far.

V. OBSERVATIONS

From the preceding calculations, it can be seen that Polynomial Regression outperforms other models for the proposed MLHIPS giving an accuracy of 80.97%. The Polynomial Regression in contrast to other models fits a curve to the dataset which increases the variance of the model thereby reducing the residual error. With an RMSE of 5100.53 and R2 value of 0.80, the model achieves superior results when using Polynomial Regression. If there exists a non-linear relationship between the target and set of predictor variables then polynomial regression eventually produces better results. Besides, Polynomial Regression Ridge and Lasso Regression of MLHIPS have achieved an accuracy value of 75.82% and 75.86%, respectively. The 6 independent parameters have a considerable correlation amongst them as a result of which lasso regression and ridge regression produces similar results. The Multiple linear regression model of MLHIPS has achieved an accuracy of 75.86%. whereas the simple linear regression had an accuracy of 62.86% which was the lowest among all. In the above scenario the dataset is partitioned into 80-20 ratio for training and testing purpose. In the dataset of 70:30 ratio, the accuracy of the polynomial

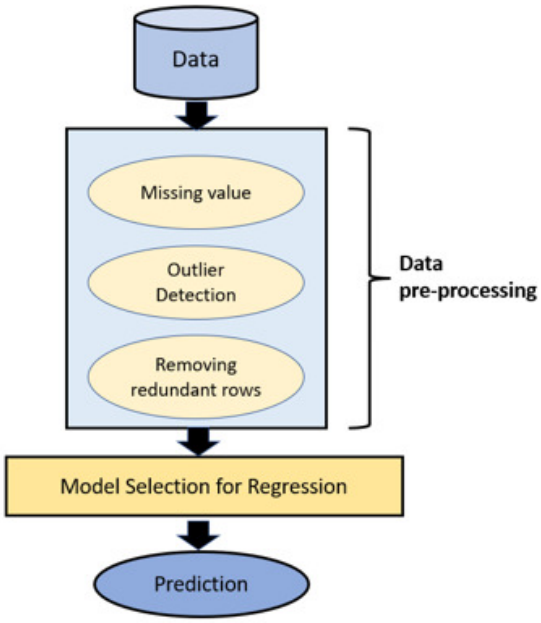

Fig.1 Flowchart of MLHIPS Model

regression decreases from previous value of 80.97% to 80.54% which is negligible, and similarly the rest of the models have also produced a drop in the accuracy values.

The multiple linear regression, ridge regression and lasso regression accuracy values are reduced by approximately 2% units from their past values of 75.86% to 73.32% for multiple linear regression, 75.82% to 73.56% for ridge regression and from 75.86% to 74.12% for lasso regression.

Further it was noted that on increasing the degree of the polynomial regression from n=2 to n=3 the accuracy has increased from previous value of 80.97% to 83.62%. But later on, further increasing the degree to 4 and higher values there was a drop in accuracy from 83.62% to 68.06% for degree n=4 and 51.98% for degree n=5. Thus, with degree n=3 the polynomial regression gives us a good accuracy in predicting the charges. A working of the MLHIPS model is shown in Fig.1.

VI. FUTURE WORK

In this paper we discussed some of the traditional regression models for our proposed problem statement, moving forward some of the other techniques like Support Vector Machine (SVM), XGBoost, Decision Tree (CART), Random Forest Classifier and Stochastic Gradient Boosting needs to be addressed as the future work. Several optimization

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techniques such as the Genetic Algorithm or the Gradient Descent Algorithm maybe applied on top of model evaluation. We can also apply some feature selection techniques to our dataset before we train our model to gain a good accuracy value as some of the features may be omitted while predicting the charges. Besides a model to perform well a good balanced dataset with a greater number of observations is required which will reduce the variability of the model so in the future if, we get more data than the model can be trained well.

VII. CONCLUSION

Calculation of health insurance charges in the traditional process are a hefty task for the insurance companies. Following human intervention in the process may sometime produce faulty or inaccurate results and also when the data increases the time taken for calculation by human's increases. In scenarios like these the implementation of Machine Learning models can be very beneficial to the company. In this paper, several Machine Learning regression models are used to predict the cost of health insurance based on specific attribute values present in the dataset. The results obtained are summarized in Table II. With an RMSE of 5100.53, an R2 of 0.80, and an accuracy of 80.97 percent, Polynomial Regression is the most efficient. Based on the model's configuration parameters which are tuned during the training phase, on the basis of performance the different proposed models are arranged accordingly starting from Polynomial Regression, Lasso and Ridge Regression, Multiple Linear Regression, and Simple Linear Regression. These models can be incorporated by the companies for calculating the charges in a fast and reliant manner thereby saving the time and cost of the company. These models in the later stage of life cycle can be deployed onto the cloud platforms when the data increases integrated with high end computing resources for faster processing of real time data in short span of time interval.

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